

Ψ -Vortex:
Structure-Regularized Recurrent Learning for Latent
Thermal-Coupling Inference and Verilog-A Compact Modeling of
3D Neuromorphic Devices

Technical Rationale and Design Evolution

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Contents

Document Status	3
1 The Modeling Gap in Electronic Design Automation	3
1.1 The Limitations of the Closed-Box Paradigm	3
1.2 The Genesis of the Ψ -Vortex Framework	4
1.3 Scope and Claims Boundary	4
2 The Lineage of Failure and Progress: The Ψ-Family History	5
2.1 Ψ -NN: The Foundational Decoupling	5
2.2 Ψ -HDL: The Bridge to Hardware	5
2.3 The TinyML Deployment Branch: From Structure to Deployment Accounting	5
2.4 Ψ -xLSTM: Recurrent High-Frequency Modeling	6
2.5 State of the Art: Automation and Initialization	6
3 The Computational Bottleneck: Analysis of Recurrent Complexity	7
3.1 The Cost of Matrix Memory	7
3.2 The Physics of Random Initialization	7
3.3 The Limits of Standard Optimization	8
4 The Structural Bottleneck: The Limits of Manual Regularization	8
4.1 The Flaw of Static Regularization	8
4.2 The Burden of Expertise	8
5 Physics-Aware Initialization as a Task-Dependent Mechanism	8
5.1 The Vortex Concept	9
5.2 Mathematical Formulation of Physics-Aware Initialization	9
5.3 Automatic Symmetry Detection	9
6 The Ψ-Vortex Methodology: BIC-Inspired Structure-Regularized Compact Extraction	9
6.1 Adaptive BIC Regularization	9
6.2 Validation-Based Architecture Selection	10
7 Experimental Architecture and Implementation Details	10
7.1 Computational Platform and Environment	11
7.2 Baseline Configurations	11

8	Case Study I: High-Frequency Memristor Modeling	11
8.1	Device Physics and Synthetic Data	11
8.2	Training Behavior and the Role of Initialization	11
9	Case Study II: Latent Thermal-Coupling Inference in a 3D Neuromorphic Scenario	12
9.1	The Latent Thermal-Coupling Scenario	12
9.2	Recurrent Recovery of Effective Thermal Coupling	12
9.3	Thermal-Aware Verilog-A Generation	13
10	Ablation Studies and Component Analysis	13
10.1	Component Analysis Under the Chronological Protocol	13
10.2	Bounded Structural Evidence	14
10.3	Compression Metrics	14
11	Implications for Sustainable AI and Future EDA	14
11.1	Energy-Efficient Edge Deployment	14
11.2	The Digital Twin and System-Level Simulation	15
11.3	Conclusion: A Structure-Regularized Recurrent Compact-Modeling Workflow	15
	References	15

Document Status

This document presents the technical rationale, historical lineage, and architectural motivations underlying the Ψ -Vortex framework. It is a design and research-program overview accompanying the Ψ -Vortex manuscript accepted for publication in IEEE Access, its Supplementary Material, and the recurrence-safe public implementation.

The reported Ψ -Vortex results follow the verified public result record. The accepted manuscript and its Supplementary Material remain the authoritative sources for complete protocols, tables, figures, and statistical detail. This rationale synthesizes the conceptual evolution of the Ψ -family and explains the engineering decisions behind Ψ -Vortex. It also distinguishes the recurrent compact-modeling contribution of Ψ -Vortex from the complementary, now-published TinyML deployment and sustainability-accounting branch of the Ψ -family [30].

Canonical Recurrent Contract

The sequence interface used for recurrent teacher and Student training, RRAD, BIC training, evaluation, clustering, and batched export accepts tensors with shape

$$[\text{batch}, \text{sequence length}, \text{features}], \quad \text{sequence length} > 1.$$

Timesteps remain chronological. Shuffling is allowed only across complete trajectories or complete windows. Source identities remain disjoint across training, validation, and test splits. Recurrent state resets between independent trajectories. For truncated backpropagation through time (TBPTT), state propagates and is then detached only between contiguous chunks from the same trajectory. Validation sources select architecture and cluster count; test sources are reserved for final reporting. Mean squared error (MSE) and the BIC-inspired score both count scalar target elements. The separate deployment `step()` interface processes one observation at a time while carrying recurrent state. It implements streaming recurrence, not a length-one replacement for sequence training or independent pointwise prediction.

1 The Modeling Gap in Electronic Design Automation

The integration of machine learning into scientific computing has significantly changed device modeling, yet the transition from methodological novelty to industrial utility remains limited by persistent inefficiencies. Within Electronic Design Automation (EDA), the challenge is not merely to approximate device behavior, but to do so under declared physical constraints with computational tractability, structural interpretability, and simulator-compatible numerical behavior. The Ψ -Vortex framework was developed to address the tension between the temporal fidelity required for history-dependent electronic systems and the compactness required for practical deployment.

1.1 The Limitations of the Closed-Box Paradigm

Historically, the application of neural networks to physical systems operated under a "closed-box" paradigm. In this mode, deep learning models functioned as universal approximators, relying solely on massive datasets to interpolate input-output relationships. While effective for static image recognition or natural language processing, this approach proved inadequate for physical device modeling. A closed-box model, lacking any internal representation of conservation laws or thermodynamic constraints, can generate non-passive or non-convergent behavior when extrapolated beyond the training regime, leading to violations that prevent stable SPICE simulation. In the context of EDA, where circuit simulators such as SPICE require passivity, continuity, and numerical stability for convergence, such behavior renders a model unsuitable for practical circuit simulation [1], [2].

The advent of Physics-Informed Neural Networks (PINNs) sought to rectify this by embedding governing differential equations directly into the loss function [3]. This ensured that the network's output respected physical laws. However, traditional PINNs introduced their own set of limitations. Standard Multi-Layer Perceptron (MLP) backbones can exhibit "spectral bias", a documented tendency to prioritize low-frequency functions while smoothing some high-frequency transients relevant to switching-device modeling [4], [5]. Furthermore, although the output of a PINN may satisfy physical constraints, the internal structure of the network typically remains a dense and uninterpretable set of parameters, offering little insight into the topology of the system being modeled. This lack of structural alignment can require

large networks even when the underlying physical dynamics admit a more compact representation, leading to inefficiency.

This efficiency crisis has spurred a wave of recent 2024-2025 innovations attempting to bridge the gap. For instance, Physics-Informed Kolmogorov-Arnold Networks (PIKANs) have emerged as an alternative backbone, leveraging learnable activation functions to better capture high-frequency dynamics [6], [7]. Simultaneously, automated frameworks like Lang-PINN have begun using multi-agent Large Language Models to synthesize PINN code directly from natural language descriptions [8]. However, while these advancements address component-level issues, they have not yet demonstrated the unified, hardware-aware structural optimization required for generating industrial-grade compact models. We provide head-to-head empirical comparisons against a Kolmogorov-Arnold network (PIKAN) and sparse identification (SINDy), in addition to an MLP, on shared synthetic and measured benchmarks; DiffPINN and LeIn-PINN, as initialization methods, are discussed only architecturally.

1.2 The Genesis of the Ψ -Vortex Framework

The Ψ -Vortex framework was developed as a direct response to specific bottlenecks that emerged within the Ψ -family lineage of Physics Structure-Informed Neural Networks. The primary research question "Why was the Ψ -Vortex framework developed?" can be answered by identifying the two critical bottlenecks that halted the progress of its predecessors:

1. **The Computational Bottleneck:** Matrix-memory recurrence introduces an $O(d^2)$ cost per timestep in the memory dimension d , making high-capacity chronological training expensive [9].
2. **The Manual Tuning Bottleneck:** The reliance on human expertise to manually specify structural hyperparameters (cluster counts, matrix ranks) prevented the true automation of model generation.

Ψ -Vortex addresses these barriers separately. Recurrent modeling addresses latent temporal coupling, BIC-inspired regularization reduces manual structure selection, and physics-aware initialization is evaluated as a task-dependent mechanism. It records a large shared-target convergence speedup on printed sequences but worse predictive fidelity on thermal recovery, so it is not treated as a universal improvement. Success is evaluated through three simultaneous criteria: (i) fidelity and interpretability of inferred states within the declared generator, (ii) numerical agreement in compiled circuit execution, and (iii) structural compactness with clearly separated effective-value, expanded-parameter, and serialized-artifact accounting. The pipeline combines recurrent temporal integration, BIC-inspired structure selection, validation-only clustering, and Verilog-A compact-model export. The architecture search space, BIC bandwidth, candidate cluster counts, and training budget remain user-specified.

1.3 Scope and Claims Boundary

The Ψ -Vortex framework is designed as a modeling and inference methodology rather than a replacement for first-principles physical solvers. The following scope boundaries define its intended use:

- Ψ -Vortex does not aim to replace TCAD or continuum thermal solvers; instead, it generates behavioral compact models suitable for circuit- and system-level simulation.
- Discovered parameters (e.g., thermal coupling coefficients) represent effective system-level descriptors rather than universal material constants.
- "Non-destructive inference" refers to inference from electrical observables without embedded sensors, not direct measurement of internal temperatures. Compact-model extraction is additionally validated on an independent public dataset of measured memristors (held-out and cross-device) [29]; the 3D thermal-coupling case study is a proof-of-concept on synthetic data calibrated to published device parameters, and experimental validation of the inferred thermal coupling on fabricated 3D stacks remains future work. Separately, the exported compact thermal-coupling model is validated through behavioral execution, native ngspice transient simulation, and compiled Verilog-A execution via OpenVAF-to-OSDI in ngspice.
- The framework targets automation and deployability within EDA workflows, not closed-form identification of underlying PDEs.
- Latent states inferred by the model may aggregate multiple slow physical processes (thermal diffusion, ionic drift, aging), and should be interpreted accordingly.

These boundaries clarify that Ψ -Vortex should be understood as a complementary modeling layer positioned between device physics and system-level simulation. The objective is therefore not to replace first-principles solvers, but to generate simulator-stable surrogate compact models that remain structurally interpretable and editable within engineering workflows. The emphasis is on repeatable structure extraction and exportability rather than purely minimizing curve-fitting error.

2 The Lineage of Failure and Progress: The Ψ -Family History

To fully appreciate the architectural decisions behind Ψ -Vortex, one must conduct a forensic analysis of the methodologies that preceded it. The Ψ -family represents a progressive sequence of methods in which each iteration solved one modeling problem while exposing a deeper engineering limitation.

2.1 Ψ -NN: The Foundational Decoupling

The journey began with the Physics Structure-Informed Neural Network (Ψ -NN) [10]. The "Why" behind Ψ -NN was the necessity to break the closed-box nature of standard PINNs. The core innovation of Ψ -NN was the principle of "Decoupled Regularization". Standard PINNs attempt to force a single network to satisfy both data constraints and physical laws simultaneously, often leading to optimization conflicts [3], [5]. Ψ -NN split this burden between two distinct entities:

- **The Teacher Network:** A high-capacity model dedicated solely to satisfying the Partial Differential Equation (PDE) residuals ($F(u) = 0$).
- **The Student Network:** A compressed model trained via knowledge distillation to mimic the Teacher while being subjected to structural regularization.

This decoupling allowed for the extraction of internal symmetries from network weights, showing that neural networks could, in principle, learn structural regularities associated with physics. However, Ψ -NN relied on the standard MLP backbone. Consequently, it retained the MLP's spectral-bias tendency, reducing its suitability for sharp, high-frequency transients observed in modern semiconductor devices. It was effective for steady-state behavior but could lose fidelity on rapid transients.

2.2 Ψ -HDL: The Bridge to Hardware

The subsequent iteration, Ψ -HDL (Physics Structured-Informed Neural Networks for Hardware Description Language Generation), addressed the practical gap between machine learning research and industrial EDA tools [11]. The motivation for Ψ -HDL was to translate the abstract weight matrices of Ψ -NN into executable behavioral Verilog-A code, the standard language for circuit simulation [12], [1].

Ψ -HDL introduced a rigorous pipeline for converting discovered weight clusters into compact Verilog-A models and established the bridge from structure discovery to simulator-oriented deployment. It did not, however, derive the full implementation-level deployment-accounting quantities later examined in the dedicated TinyML study [30].

- **The Remaining Limitation:** Despite its success in hardware translation, Ψ -HDL was still built upon the MLP backbone and could therefore retain the spectral-bias tendency of its predecessor. It could generate Verilog-A models, but those models were less well suited to high-frequency regimes in which smoothing can reduce the fidelity of rapid switching dynamics.

2.3 The TinyML Deployment Branch: From Structure to Deployment Accounting

A complementary branch of the Ψ -family investigated what happens after a physics-structured model leaves the modeling environment and enters the TinyML deployment pipeline [30]. Rather than focusing on compact-model generation, this work examined whether architecture-encoded physical structure survives clustering, finite-precision quantization, compilation, and microcontroller execution, and how the resulting representation should be accounted for at the hardware level.

For the evaluated Burgers surrogate, architecture-encoded antisymmetry was tracked through clustering, INT8 quantization, compilation, and Cortex-M target-ISA emulation. The study also demonstrated that structural compactness and implementation compactness are distinct quantities: centroid count alone

does not represent deployable memory because the relation/index representation must also be stored. Flash, statically allocated RAM, executed instructions, and firmware numerical outputs were therefore evaluated separately.

The work additionally examined training-budget sensitivity under intermittent-power scenarios and separated reduced effective training budget from rollback and checkpoint-state degradation. Its sustainability analysis likewise separated model-structure effects from system-level hardware substitution. In the evaluated lifecycle scenario, the approximately $45\times$ five-year carbon ratio was dominated by GPU-to-microcontroller hardware substitution rather than by the structured representation itself.

The Remaining Limitation: This deployment-oriented study established how physics structure can be followed through TinyML compilation, execution, and resource accounting, but it did not address chronological latent-state inference or the extraction of recurrent simulator-executable compact models for history-dependent devices. Those requirements motivate the recurrent Ψ -xLSTM and Ψ -Vortex branches described below.

2.4 Ψ -xLSTM: Recurrent High-Frequency Modeling

The recurrent predecessor, Ψ -xLSTM, was developed to mitigate the spectral limitations of an MLP backbone [13]. It introduced Extended Long Short-Term Memory (xLSTM) cells with exponential gating and matrix memory [9]. These mechanisms increase the capacity to represent rapid temporal variations and motivated recurrent modeling of sharp transient and high-frequency device dynamics. Their demonstrated benefit is bounded to the tested regimes and does not imply universal elimination of spectral bias. The present Ψ -Vortex evidence is established independently under true chronological sequence training. A matched vanilla LSTM succeeds in 14 of 15 cells across the five-coupling sweep, so the evidence identifies recurrence as the decisive class property without claiming that one Ψ -specific recurrent cell is uniquely necessary.

To support this, Ψ -xLSTM introduced Recurrent Relation-Aware Distillation (RRAD). The implementation used here matches Student and Teacher outputs, output time differences, recurrent hidden trajectories after learned projection, and hidden-state time differences. The ablation results show that this improves predictive fidelity, but it is not required for nominal coupling acceptance.

The Emergence of the Crisis:

Matrix-memory recurrence introduces a computational bottleneck alongside its temporal representation capacity. In the canonical Ψ -Vortex Teacher, the memory update is

$$C_t = \bar{f}_t C_{t-1} + \bar{i}_t (v_t k_t^\top).$$

Here $\bar{f}_t$ and $\bar{i}_t$ are scalar feature means of the forget and input gates, respectively, and k_t is the normalized key. This update involves an outer product between value v_t and key k_t vectors, resulting in a matrix memory $C_t \in \mathbb{R}^{d \times d}$, consistent with Equation (3) of the accepted manuscript. The corresponding per-timestep matrix-memory cost is quadratic in d , not in the number of training samples.

Complexity clarification. Here, the quadratic term refers to the dimensionality of the matrix memory, $d \times d$, not the dataset size. The dominant matrix-memory operation costs $O(d^2)$ per timestep, so total work accumulates across the temporal rollout, batches, and epochs. This dependence creates a practical bottleneck in high-capacity Teacher models. Physics-aware initialization was therefore evaluated as a possible way to reduce the number of training epochs. Its benefit proved task-dependent rather than universal.

Furthermore, Ψ -xLSTM, like Ψ -NN and Ψ -HDL before it, relied on manual structure extraction. A human expert was required to inspect the singular value spectra of the weight matrices and manually select the rank (r) and cluster count (K). This "Manual Tuning Bottleneck" meant the framework could not be deployed as a substantially automated tool.

2.5 State of the Art: Automation and Initialization

Recent work has explored several different approaches to these bottlenecks, providing useful context for the design choices behind Ψ -Vortex.

- **Automated Frameworks:** The Lang-PINN framework recently introduced a multi-agent approach where Large Language Models (LLMs) parse natural language descriptions to generate executable PINN code [8]. While this solves the "usability" barrier for non-experts, it focuses on code generation rather than the discovery of the underlying physical structure.
- **Modern Compact Modeling:** In the specific domain of RRAM modeling, the authors in [14] proposed an ML-driven approach to generate Verilog-A models that account for variability. However, their method relies on deep ensemble techniques and manual state determination functions, contrasting with the substantially automated, information-theoretic structure discovery proposed by Ψ -Vortex.
- **Initialization Strategies:** DiffPINN [15] and LeIn-PINN [16] address PINN convergence through generative or learned initialization. Ψ -Vortex instead evaluates an analytic symmetry and retention prior that needs no separate pre-training. On the declared shared-target printed-sequence task it reaches the target in 6.8 epochs on average versus 462 for random initialization, a recorded 116.6-fold speedup. On thermal recovery, however, random initialization gives lower test MSE, 0.02467 versus 0.17526. No direct head-to-head experiment with DiffPINN or LeIn-PINN was conducted, so only an architectural comparison is made.
- **Information-Theoretic Goals:** Parallel research on "infomorphic" networks explores local information-theoretic goal functions for neural learning [17]. It provides context for interpretable learning objectives, but does not derive or establish classical BIC guarantees for the differentiable structural surrogate used by Ψ -Vortex.

The Thesis of Ψ -Vortex:

Ψ -Vortex was developed to retain recurrent temporal integration while (i) reducing manual structure-selection effort within declared bounds and (ii) enabling latent thermal-coupling inference from electrical observations in a calibrated synthetic proof-of-concept. It does not autonomously discover arbitrary multiphysics mechanisms. The target physical setup, observables, bandwidth, candidate structures, and search space remain user-specified. Physics-aware initialization is retained as a task-dependent ablation rather than a universal mechanism. The central goal is to combine recurrent latent-state inference, structure-regularized compact extraction, and Verilog-A compact-model execution within one reproducible workflow.

3 The Computational Bottleneck: Analysis of Recurrent Complexity

This section describes the computational motivation for physics-aware initialization. The final evidence is task-dependent: the prior accelerates one shared-target printed-sequence task, but it does not improve thermal-recovery fidelity and is therefore not presented as a universal remedy for recurrent training cost.

The first major design objective of Ψ -Vortex was to address the computational bottleneck introduced by recurrent high-fidelity modeling. To understand the severity of this problem, one must analyze the interaction between the xLSTM architecture and the optimization landscape of physics-informed loss functions.

3.1 The Cost of Matrix Memory

The xLSTM architecture expands a conventional vector state with matrix-memory operations. This increases representational capacity for relational dynamics but also increases cost. The term $v_t k_t^\top$ is an outer product that scales with the square of the memory dimension. In a high-capacity teacher, this dimension can be large, for example $d = 64$ or $d = 128$.

In chronological training, the $O(d^2)$ operation is repeated for each timestep, trajectory batch, and epoch. Long sequences therefore accumulate substantial floating-point work even though the quadratic variable is the memory dimension rather than the dataset size.

3.2 The Physics of Random Initialization

The inefficiency is compounded by the "starting point problem". In conventional deep learning, weights are typically initialized using variance-preserving random schemes, such as Xavier/Glorot [18] or He initialization [19], to stabilize signal propagation across layers. From a physical modeling perspective, a randomly initialized network represents a parameter configuration that does not satisfy known symmetries or conservation constraints of the target system. It encodes no known conservation structure, no explicit symmetry, and no physically motivated prior state.

When a physics-informed loss is applied to such a network, the optimizer must move from a generic starting point toward a region that satisfies the data and imposed physical constraints. This motivated the hypothesis that a physics-consistent initialization could shorten convergence. The final experiments show that the hypothesis is task-dependent: it is strongly supported by the declared shared-target printed-sequence experiment but does not translate into improved thermal-recovery fidelity.

3.3 The Limits of Standard Optimization

Standard optimizers such as Adam or L-BFGS are general-purpose tools and have no inherent awareness of a target symmetry. When an odd relation such as $I(-V) = -I(V)$ is justified, an analytic prior can encode it at initialization. The automatic detector must nevertheless be allowed to abstain because measured or multi-input behavior need not exhibit global odd symmetry. In the final experiment it abstains with `none` on all five seeds, and the automatic and identity modes have the same validation MSE. This is valid fallback behavior rather than evidence that odd symmetry was recovered.

4 The Structural Bottleneck: The Limits of Manual Regularization

The second driver for the development of Ψ -Vortex was the "Manual Tuning Bottleneck." While previous frameworks established the possibility of structure extraction, they failed to automate it.

4.1 The Flaw of Static Regularization

Ψ -HDL employed a static L2 regularization term in its loss function, $\lambda \sum \|\theta_S\|^2$. This penalty encourages weights to be small, driving them toward zero. However, "small" is not the same as "clustered" or "structured". To extract a compact model from an L2-regularized network, a human engineer must perform a post-hoc analysis. They must visualize the weight histogram and choose a cutoff threshold ϵ . Weights below the threshold are pruned; weights within a declared range are grouped.

This process is sensitive to threshold choice and expert judgment. An overly large ϵ may remove useful structure, whereas an overly small ϵ may retain redundancy. The number of weight clusters K is a structural hyperparameter requiring a selection rule. It counts shared centroid values, not physical components in an equivalent circuit; no one-to-one mapping between weight clusters and physical circuit elements is established.

4.2 The Burden of Expertise

This reliance on manual tuning creates a bottleneck where the speed of model generation is limited by the availability of human domain experts. In an industrial EDA context, where thousands of devices might need to be modeled for a new process design kit (PDK), requiring a PhD-level expert to manually tune the hyperparameters for every single device is unscalable.

Manual selection commonly favors conservative capacity to avoid under-parameterization. Ψ -Vortex replaces a single expert guess with validation-based comparison across a bounded candidate set. It does not claim a mathematically minimal architecture or remove the need to define the candidate set, bandwidth, and training budget.

5 Physics-Aware Initialization as a Task-Dependent Mechanism

To investigate the computational bottleneck, Ψ -Vortex introduced a physics-aware initialization mechanism. The construction embeds a declared symmetry or retention prior before training. Its effect is assessed rather than assumed: it accelerates the shared-target printed-sequence task but gives worse thermal-recovery MSE than random initialization.

5.1 The Vortex Concept

The term " Ψ -Vortex" draws by analogy from the dynamical-systems concept of a basin of attraction, namely the set of initial conditions whose trajectories converge toward a given attractor [21], [22]. The analogy is informal; no formal proof that the initialization lies within a specific basin of the loss landscape is provided. The tested mechanism applies a declared or data-selected transform to eligible input and output weights and separately sets recurrent forget biases to a physical retention prior.

5.2 Mathematical Formulation of Physics-Aware Initialization

For each eligible nonrecurrent weight matrix, the implemented transform is

$$\theta_{\text{init}}^{(l)} = s \left[T_g \left(W_{\text{orth}}^{(l)} \right) + \epsilon \sigma Z^{(l)} \right], \quad Z^{(l)} \sim \mathcal{N}(0, I).$$

Here T_g is the identity for $g = \text{none}$, applies the declared sign mask to the selected input feature for $g = \text{odd}$, and takes the elementwise absolute value for $g = \text{even}$. The implementation uses $s = 0.1$, $\epsilon = 0.01$, and $\sigma = 0.01$, with eligible biases initialized to 0.01. When recurrent preservation is enabled, recurrent matrices remain orthogonal and bypass this transform. The transform is an initialization bias, not a permanent weight-tying constraint. The recurrent forget bias separately encodes retention near $\exp(-\Delta t / \tau_{\text{th}})$.

5.3 Automatic Symmetry Detection

A paired-reflection detector tests whether the data support odd or even behavior for a selected input feature. It can also abstain, so the user is not forced to impose a symmetry that the observations do not support.

The algorithm analyzes input-output pairs (X, Y) and calculates correlation coefficients for different symmetry hypotheses:

- $C_{\text{odd}} = \text{Corr}(Y(X), -Y(-X))$
- $C_{\text{even}} = \text{Corr}(Y(X), Y(-X))$

The detector selects the highest supported score only when the declared confidence and pairing requirements are met; otherwise it returns $g = \text{none}$.

This mechanism allows the framework to choose among odd, even, and neutral initialization when the data support a tested symmetry. If no hypothesis exceeds the predefined confidence threshold, the detector abstains and uses neutral initialization. On the final five-seed experiment it abstains with **none** in every seed. Automatic and identity initialization both give validation MSE 0.02513, whereas imposed odd symmetry gives 0.03544. The result supports safe fallback, not discovery of odd symmetry. Extension to mixed, weak, conditional, or axis-specific symmetries remains future work.

6 The Ψ -Vortex Methodology: BIC-Inspired Structure-Regularized Compact Extraction

To reduce the structural bottleneck, Ψ -Vortex combines a differentiable BIC-inspired structural surrogate with validation-only selection from declared candidates. It reduces manual extraction without eliminating user-specified search bounds.

6.1 Adaptive BIC Regularization

The central idea is to embed a differentiable BIC-inspired score into the loss function [23]. Classical BIC is defined as $k \ln(n) - 2 \ln(\hat{L})$, where k is an integer parameter count and $\hat{L}$ is a maximized likelihood. The score used here replaces these terms with supervised MSE and a smooth local-density estimate of effective degrees of freedom. In every multi-output path, n is the number of scalar target elements, exactly matching the denominator used by MSE. No classical BIC model-selection consistency guarantee is claimed.

Ψ -Vortex introduces a differentiable approximation of k using a local kernel-density estimator. The Student objective is

$$\mathcal{L}_{\text{Vortex}}(\theta_S) = \mathcal{L}_{\text{sup}}(\theta_S) + \mathcal{L}_{\text{RRAD}}(\theta_S) + \lambda_{\text{struct}} \mathcal{R}_{\text{BIC}}(\theta_S, E, n).$$

The supervised term is the element-mean squared error against the target trajectories. RRAD separately matches Teacher and Student outputs, projected hidden trajectories, and their time differences. These terms have distinct roles, as in Equation (7) of the accepted manuscript.

The differentiable BIC-inspired score estimates effective degrees of freedom by measuring local weight density:

$$D_i(\theta) = \sum_{j=1}^W \exp\left[-\frac{(\theta_i - \theta_j)^2}{2\gamma^2}\right], \quad d_{\text{eff}}(\theta) = \sum_{i=1}^W \frac{1}{\max(D_i(\theta), \varepsilon_D)},$$

$$\mathcal{R}_{\text{BIC}}(\theta, E, n) = n \ln(\max(E, \varepsilon_M)) + \ln(n) d_{\text{eff}}(\theta),$$

where E is element-mean MSE and n is the number of scalar target elements.

Computational feasibility. All reported training and validation-selection scores use the exact blocked expression with $\gamma = 0.1$. Exact value and gradient comparisons are available through 8,000 weights. Mini-batch and random-Fourier-feature approximations are evaluated only as scalability studies and execute at 50,000 weights; exact agreement is not claimed above 8,000. The mini-batch value error is at most 1.9% in the compared regime but its gradient cosine falls to 0.462, while the random-Fourier approximation has about 7.8% value error and gradient cosine of at least 0.946.

Hyperparameter scope. The final configuration fixes $\gamma = 0.1$. A joint trained-weight bandwidth sweep is outside the scope of the study, so new device classes must validate the bandwidth and cluster candidates on held-out trajectories.

Mechanism of Action:

The nested summation term calculates the density around each weight w_i .

- If a group of M weights clusters tightly together (approaching a single shared value), the inner sum approaches M .
- The contribution of this cluster to the total count is roughly $M \times (1/M) = 1$.
- Thus, a cluster of M weights counts as only 1 effective parameter.

This formulation makes the effective-count surrogate a smooth, differentiable function of the weights. During training, it encourages shared or nearby parameter values. After training, the pipeline materializes each declared cluster-count candidate and selects it using validation BIC. Low-rank Student ranks are supplied explicitly or evaluated as declared architecture candidates; they are not inferred by the differentiable BIC term.

6.2 Validation-Based Architecture Selection

Macro-architectural hyperparameters, including hidden dimension h , layers L , and memory size m , are selected by bounded validation comparison. Cluster count is selected separately from declared candidates, and low-rank candidates remain explicit inputs.

$$a^* = \operatorname{argmin}_{a \in A} \text{MSE}_{\text{val}}(a)$$

The search space is $A = \{h \in \{16, 32, 64, 128\}, L \in \{1, 2, 3\}, m \in \{8, 16, 32\}\}$. The algorithm trains the 36 bounded candidates and selects the minimum validation MSE without consulting the test set.

The bounded search lowers mean validation MSE from 0.02539 for the declared reference configuration (64, 2, 16) to 0.01911, a 24.7% improvement. Five seeds select five different configurations, so no universal architecture is claimed. This is grid evaluation over declared candidates, not an unconstrained neural architecture search.

7 Experimental Architecture and Implementation Details

To rigorously validate the claims of automation, structure discovery, and latent-coupling recovery, the Ψ -Vortex framework was subjected to a comprehensive experimental evaluation. The setup was designed to ensure strict reproducibility and to isolate the contributions of specific mechanisms.

7.1 Computational Platform and Environment

The verified computational record uses the following workstation and software environment.

- **GPU:** NVIDIA RTX 4090 with 24 GB video memory.
- **CPU:** Intel Core i9-13900K.
- **Memory:** 128 GB of RAM.
- **Software Stack:** Windows 11, Python 3.14, PyTorch 2.9.0+cu130, NumPy 2.3.4, and CUDA acceleration.

Every run records its configuration, package version or commit, random seed, device, source identities by split, architecture, selected cluster count where applicable, checkpoint hash, metrics, runtime, and generated artifacts. All declared seeds are retained; no best-seed selection is used.

7.2 Baseline Configurations

The reference architectures provide methodological context, while the matched head-to-head experiments use separately specified baselines. Their roles are distinguished below:

1. **Reference PINN:** A four-layer, 64-unit tanh MLP from the printed-memristor modeling reference configuration [25]. This reference architecture is distinct from the static MLPs used in the matched head-to-head experiments. Those MLPs have two 32-unit tanh hidden layers; the measured-data model has two inputs and 1,185 parameters, whereas the synthetic full-sweep model has one input and 1,153 parameters. The separate nominal pointwise recurrence control uses two 16-unit hidden layers.
2. **Recurrent Baselines:** The direct Ψ -xLSTM predecessor motivates the matrix-memory Teacher and RRAD. A chronological vanilla LSTM is also evaluated on the same five-coupling sweep and succeeds in 14/15 cells, isolating recurrence from claims about a uniquely necessary cell.
3. **Contemporary Architecture Baseline (PIKAN):** The public package implements a Chebyshev-basis Kolmogorov–Arnold network, not the third-party `pykan` implementation. The matched experiments use two Chebyshev layers with an eight-unit hidden width and polynomial degree three. It is evaluated as a static alternative-backbone baseline on the same source-disjoint splits.
4. **Classical Identification Baseline (SINDy):** Sparse identification of nonlinear dynamics (sequentially thresholded least squares over a polynomial library), included as an interpretable, closed-form baseline for both compact modeling and latent-coupling recovery.

The head-to-head static and recurrent baselines use matched source-disjoint splits. The reference PINN description does not identify the architecture used for those separate result rows. Pointwise, state-reset, shuffled-order, and absent-memory variants are experimental controls only and are not normal recurrent execution paths.

8 Case Study I: Printed-Memristor Sequence Modeling

Rapid device transients and the spectral-bias literature motivate recurrent modeling [4], [5]. The reported initialization experiment specifically evaluates convergence on archived, chronologically ordered printed-memristor sequences. It is not a direct measurement of spectral bias or a demonstration of its elimination.

8.1 Device Physics and Synthetic Data

The printed-memristor modeling context is motivated by flexographically printed Ag/PMMA:PVA/ITO devices [24], [25]. The convergence experiment reads the archived synthetic current–voltage sequences, groups them by complete device and cycle identities, and keeps source devices disjoint across training, validation, and test partitions.

- **Inputs and target:** The recurrent inputs are the archived voltage and normalized position within each cycle; the target is the scaled current. The loader preserves row order within each cycle and uses sample indices as its time coordinate. It does not establish a physical 150 kHz excitation for this convergence experiment.
- **Chronological protocol:** Complete cycles retain their ordering, and recurrent state resets between independent trajectories. Both initialization modes use the same declared training budget and source-level splits.
- **Objective:** For each of five declared seeds, the shared target is the minimum recorded training loss of that seed’s random-initialization baseline over the complete budget. Both modes are scored by their first epoch reaching that same target. This is a retrospective, per-seed convergence comparison, not a fixed 10^{-6} threshold or an early-stopping runtime experiment.

8.2 Training Behavior and the Role of Initialization

Under the declared shared-target printed-sequence protocol, physics-aware initialization reaches the common target in 6.8 epochs on average, compared with 462 epochs for random initialization. The mean per-seed speedup is 116.6-fold. This convergence result does not imply universally better fidelity. On thermal recovery, random initialization gives test MSE 0.02467, whereas physical-retention initialization gives 0.17526.

The two findings address different tasks and must be reported together. Initialization can accelerate a declared convergence target without guaranteeing better transfer or final fidelity. Latent-coupling recovery is enabled by chronological recurrence, while the BIC-inspired mechanism improves the tested accuracy–compactness trade-off. The evidence does not prove an initialization-independent unique optimum.

9 Case Study II: Latent Thermal-Coupling Inference in a 3D Neuromorphic Scenario

The second case study was designed to demonstrate the latent-variable inference capabilities of the framework. It addressed the "3D Neuromorphic Grand Challenge", specifically the problem of invisible thermal crosstalk in massive vertical stacks.

The motivation for this experiment extends beyond neuromorphic demonstration. Recent reviews of advanced 3D integrated systems have identified thermal transport through interlayer dielectrics and heterogeneous interfaces as a primary bottleneck to continued scaling, particularly for AI and high-density computing [26], [27]. In such architectures, localized hotspots and inter-layer thermal coupling are difficult or impossible to observe through direct, non-destructive metrology. As a result, thermal behavior must often be inferred indirectly rather than measured.

The following case study examines whether such hidden coupling mechanisms can be detected and parameterized solely from electrical behavior, without explicit thermal sensors or predefined heat-transport equations. This case study is conducted on synthetic data calibrated to published device parameters; experimental validation on fabricated 3D stacks remains future work.

9.1 The Latent Thermal-Coupling Scenario

The 1000-layer printed-conductor architecture reported by Strutwolf et al. provides motivation for investigating interlayer interactions [28]. The evaluated latent-recovery benchmark is a calibrated synthetic driver-victim pair across adjacent layers, not a simulation or measurement of a complete 1000-layer memristor stack.

- **Topology:** A "Driver" memristor on Layer N and a "Victim" memristor on Layer N+1, separated by a 100 μm paper substrate.
- **The Phenomenon:** Heat generated by the Driver diffuses through the substrate and affects the conductivity of the Victim. This is a "Ghost Coupling" effect where the Victim's current changes even though its voltage is constant.
- **Benchmark assumption:** A fixed, low victim read bias motivates isolating thermally induced changes from switching. In the implemented synthetic generator, thermal coupling is imposed through the normalized log-current target $\alpha\Delta T$. Isolation of that mechanism is therefore a modeling assumption, not an experimentally established exclusion of electrical switching or other mechanisms in fabricated devices.

9.2 Recurrent Recovery of Effective Thermal Coupling

The recurrent predictor is trained on chronological driver-voltage trajectories and the synthetic normalized log-current response. Temperature is not supplied as a predictor input. Post-hoc coefficient evaluation does use the generator's temperature-rise trajectory ΔT as the regressor against which the predicted response is fitted. The recovered coefficient is thus assessed within the generator-matched benchmark, not through independent thermometry or unconstrained identification of a fabricated buried interface.

On the synthetic benchmark, the framework inferred a thermally meaningful latent coupling state from electrical observations alone.

1. **Detection:** The xLSTM architecture detected a slow-moving drift in the Victim's current that could not be explained by the fast electrical inputs.
2. **Allocation:** The BIC-inspired mechanism allocates compact recurrent capacity to the latent state. The numerical coupling coefficient is recovered afterward on held-out driver trajectories using a free-intercept least-squares estimator and is not directly output by BIC. The readout reports a value only when held-out R^2 reaches 0.8; otherwise it abstains.
3. **Reported evidence:** Figure 5 of the accepted manuscript summarizes coefficient recovery and held-out acceptance across the five-coupling sweep; Figure 4 reports the recurrence controls. Figure 3 instead presents structural-selection ablations. These results assess effective coupling and predictive fidelity, not direct reconstruction or measurement of substrate temperature in a fabricated stack.
4. At the nominal $\alpha = 0.08$, the three-seed mean is $\hat{\alpha} = 0.08738 \pm 0.00900$, held-out $R^2 = 0.981$, with 3/3 accepted seeds.

The supported operating regime is bounded. At $\alpha = 0.05$ and 0.08, all three seeds pass, with mean relative errors of 4.6% and 9.2%. At $\alpha = 0.10$, 0.15, and 0.20, only 1/3, 1/3, and 0/3 seeds pass, with mean relative errors of 36.0%, 56.5%, and 58.6%. State-reset, shuffled-order, and absent-memory pointwise controls accept 0/9 at the nominal condition; their largest mean R^2 is 0.009. Static MLP, PIKAN, and SINDy controls accept 0/15 each across the sweep, while a chronological vanilla LSTM accepts 14/15. The conclusion is therefore that chronological recurrence is necessary for this benchmark, not that the Ψ -specific cell is uniquely necessary or that full-range recovery has been established. The inferred coefficient remains an effective descriptor of the generator-matched law rather than a universal material constant.

9.3 Thermal-Aware Verilog-A Generation

The validated Verilog-A artifact is a sampled-state export of the selected one-layer clustered gated recurrent unit (GRU). It exposes `in_0` and `in_1` as feature inputs, `sample_clock` as a two-phase update control, and `out_0` as the learned output. Internal capacitor-backed state and candidate nodes implement recurrent state. The generated source expands every affine product and learned constant componentwise in valid Verilog-A [1], [12].

The exported recurrent map is

$$\begin{aligned} r_t &= \sigma(W_{ir}x_t + b_{ir} + W_{hr}h_{t-1} + b_{hr}), \\ z_t &= \sigma(W_{iz}x_t + b_{iz} + W_{hz}h_{t-1} + b_{hz}), \\ n_t &= \tanh(W_{in}x_t + b_{in} + r_t \odot (W_{hn}h_{t-1} + b_{hn})), \\ h_t &= (1 - z_t) \odot n_t + z_t \odot h_{t-1}, \\ y_t &= W_{out}h_t + b_{out}. \end{aligned}$$

During the low clock phase, each candidate node is driven toward the newly computed h_t while persistent state is held. During the high phase, persistent state is driven toward the candidate while the candidate is held. Independent transient simulations start from zero capacitor voltage, so recurrent state resets between stimuli and propagates only within one stimulus.

The reported $\hat{\alpha} = 0.08738$ is the three-seed mean recovered post hoc on disjoint held-out driver trajectories at $\alpha = 0.08$. It is not a literal parameter in the recurrent Verilog-A artifact and is not selected from a best seed.

Compiled circuit validation is distinct from TorchScript validation. Both guarded TorchScript exports reject length-one calls through the batch interface while retaining a stateful one-sample `step()` method; the largest batch-versus-streaming absolute error is 1.39475×10^{-5} . Separately, OpenVAF-reloaded 20260616-2-gc592eed6 compiles the sampled-state Verilog-A to the Open Simulator Design Interface (OSDI), and ngspice 45.2 executes all five declared stimuli. The worst OSDI-versus-PyTorch absolute error is 5.46249×10^{-5} , below the declared 10^{-3} tolerance. This validates equation-level circuit execution within the tested stimuli. It is not process design kit qualification or fabricated-device validation.

10 Ablation Studies and Component Analysis

To characterize the individual roles of initialization, recurrence, RRAD, and structural regularization, a systematic component analysis was conducted under the canonical chronological sequence protocol.

10.1 Component Analysis Under the Chronological Protocol

Under sequence-mode training, the components contribute as follows.

Physics-aware initialization is task-dependent. It records a 116.6-fold shared-target convergence speedup on printed sequences, but random initialization gives better thermal-recovery test MSE, 0.02467 versus 0.17526.

Adaptive BIC regularization improves the tested accuracy–compactness trade-off. Full BIC selects 16 effective centroid values and gives mean test MSE 0.02467. No-BIC, L1, and L2 retain 273 values with MSE near 0.062–0.065, while fixed-cluster and fixed-rank alternatives lose more fidelity.

Recurrent temporal integration enables nominal latent-coupling recovery. State-reset, shuffled-order, and pointwise controls accept 0/9, whereas chronological recurrence accepts 3/3 at $\alpha = 0.08$. Full Recurrent Relation-Aware Distillation (RRAD) gives mean test MSE 0.02467. Removing hidden matching, temporal matching, or both increases prediction error, but every RRAD arm still accepts all three nominal seeds. RRAD is therefore credited with predictive fidelity, not with being necessary for recurrence.

10.2 Bounded Structural Evidence

The structural evidence supports a bounded conclusion. BIC and validation clustering reduce manual extraction within declared candidate sets, but bandwidth, candidate clusters, architecture bounds, and training budgets remain user choices. The experiment does not prove a unique mathematical minimum or an initialization-independent universal optimum. Repeated-seed agreement should therefore be reported as empirical stability within the tested setup rather than as proof that the recovered topology is determined uniquely by physics.

10.3 Compression Metrics

The canonical synthetic teacher contains 26,273 expanded parameters. Validation-selected clustering produces an exported recurrent gated unit with 273 expanded parameters and 16 centroid values, a 98.96%

reduction in expanded parameter count. The expanded float32 parameters occupy 1,092 bytes, while the serialized TorchScript artifact occupies 14,875 bytes. These quantities describe different objects and must not be interchanged; the TorchScript file is not centroid-coded storage.

The separate measured-device compression sweep gives teacher mean normalized root mean squared error (NRMSE) 0.0156. The lowest student-architecture mean is 0.0319 at 3,489 expanded parameters. The smallest tested student contains 28 expanded parameters and has mean NRMSE 0.0660. This is an empirical compression-fidelity curve, not evidence that the smallest model is universally optimal. Quantization, fixed-point execution, large-array performance, and hardware energy remain unvalidated.

11 Implications for Sustainable AI and Future EDA

The development of Ψ -Vortex extends beyond the immediate technical metrics of speed and accuracy. It addresses broader systemic challenges in the semiconductor industry and the field of artificial intelligence.

11.1 Energy-Efficient Edge Deployment

The energy cost of training and deployment is a relevant motivation, but Ψ -Vortex itself does not measure hardware energy. Its demonstrated efficiency result is a 98.96% reduction in expanded parameter count for the canonical clustered recurrent export, together with explicit artifact-size accounting. This makes Ψ -Vortex a candidate for resource-constrained deployment, contingent on future fixed-point, throughput, memory, and energy measurements on representative hardware.

A complementary published study in the Ψ -family examined these deployment questions directly in the TinyML setting [30]. That work followed architecture-encoded physical structure through clustering, INT8 quantization, compilation, and Cortex-M target-ISA emulation and separately accounted for deployable weight memory, linked-binary flash, statically allocated RAM, executed instructions, and derived latency and energy under stated assumptions. It showed that structural compression does not automatically imply equivalent implementation-level resource reduction.

The sustainability result is similarly bounded. In the evaluated lifecycle scenario, the approximately $45\times$ five-year carbon ratio was dominated by GPU-to-microcontroller hardware substitution rather than by the physics-structured representation itself [30]. Ψ -Vortex therefore complements, rather than duplicates, this deployment-oriented work: the TinyML study establishes a methodology for following physics structure into resource-constrained execution and sustainability accounting, whereas Ψ -Vortex concentrates on recurrence-safe latent-state inference, validation-selected structural compression, and simulator-executable recurrent Verilog-A compact models.

11.2 The Digital Twin and System-Level Simulation

The ability to generate multi-port thermal-aware compact models supports executable component-level digital twins. The sampled-state Verilog-A artifact is tested beyond symbolic generation through OpenVAF-reloaded, OSDI loading, and ngspice execution. The validated result is limited to the declared pairwise synthetic regime. The multilayer extension is not established: each of the three layers accepts only 1/3 seeds, with mean coefficient errors above 50%. Scaling to distributed stacks therefore remains open and may require cumulative coupling, higher-order interactions, distributed thermal networks, and fabricated-device validation.

11.3 Conclusion: A Structure-Regularized Recurrent Compact-Modeling Workflow

In conclusion, Ψ -Vortex is a recurrence-safe, structure-regularized compact-modeling workflow for EDA. At the nominal synthetic condition $\alpha = 0.08$, it recovers $\hat{\alpha} = 0.08738$ with held-out $R^2 = 0.981$ and 3/3 accepted seeds, while all three no-memory controls accept 0/9. The broader sweep shows a clear boundary: only $\alpha = 0.05$ and 0.08 pass every seed. BIC-inspired regularization and validation-selected clustering reduce the canonical teacher from 26,273 to 273 expanded parameters with 16 centroid values. Separate measured-device experiments give held-out seed-mean correlations from 0.997 to 1.000. Five compiled Verilog-A stimuli execute in ngspice with worst absolute error 5.46249×10^{-5} , below the 10^{-3} tolerance.

These results establish a reproducible path from chronological trajectories through recurrent inference and structural compression to equation-level circuit execution within the tested pairwise regime. They do not establish universal full-range recovery, reliable multilayer recovery, process design kit qualification, or fabricated-stack thermal discovery. Physics-aware initialization is task-dependent, and a vanilla LSTM result confirms that recurrence, rather than one uniquely Ψ -specific cell, is the relevant capability. Fabricated-device populations, distributed coupling, missing-data robustness, and deployment beyond the current TorchScript path remain important future work.

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